

GIFT 1 CASE STUDY PROJECT

E-COMMERCE SALES DATA CLEANING & TRANSFORMATION

For Excel Study Group

1. PROJECT OVERVIEW

By completing this case study, you will master:

Skill Area	What You'll Learn
Folder Import	Import multiple CSV files from a folder as a single operation
Basic Cleaning	Handle nulls, duplicates, whitespace, and inconsistent formats
Text Transformations	Standardize text case, fix typos, remove special characters
Date Handling	Normalize multiple date formats into a single standard
Data Type Management	Convert text to numbers, fix currency symbols, handle parentheses
Merge & Append	Combine related tables using keys and relationships
Advanced M Code	Write custom M functions for complex transformations
Quality Assurance	Validate data integrity after transformation

Files Included in This Case Study

File Name	Description	Records
Messy_List_of_Orders.csv	Customer order headers with messy data	~508 rows
Messy_Order_Details.csv	Line-item details with data quality issues	~1,506 rows
Messy_Sales_Target.csv	Monthly sales targets with inconsistencies	~44 rows

2. DATASET DESCRIPTIONS

2.1 Messy_List_of_Orders.csv

Purpose: Contains one record per order with customer and location information.

Column	Data Type	Description	Known Issues
Order ID	Text	Unique order identifier	Leading zeros, trailing spaces, duplicates
Order Date	Text/Date	Date the order was placed	Multiple formats (DD-MM-YYYY, MM/DD/YYYY, YYYY-MM-DD, etc.)
Customer Name	Text	Name of the customer	Extra whitespace, some null values
State	Text	Indian state name	Typos, abbreviations, inconsistent casing
City	Text	City name	Mixed case (ALL CAPS, all lowercase)

Column	Data Type	Description	Known Issues
Notes	Text	Internal notes	Mostly empty, irrelevant for analysis
Sales_Rep	Text	Sales representative	Mostly null, irrelevant for analysis

● **Data Quality Issues to Fix:** - Inconsistent Date Formats: Some dates are DD-MM-YYYY, some MM/DD/YYYY, some YYYY-MM-DD, and some say “TBD” - Extra Whitespace: Leading/trailing spaces in CustomerName and Order ID - Leading Zeros: Some Order IDs have a “0” prefix (e.g., 0B-25601) - Trailing Spaces: Some Order IDs have spaces at the end - Null Values: Some CustomerName and State cells are blank or null - Inconsistent State Names: “Maharashtra” appears as “maharashtra”, “Maharastra”, “MAHARASHTRA”, “MP”, etc. - Inconsistent City Names: Some cities are in ALL CAPS or all lowercase - Duplicate Rows: Both exact duplicates and near-duplicates exist - Irrelevant Columns: Notes and Sales_Rep columns should be removed - Placeholder Values: Some cells contain “N/A” or “TBD”

2.2 Messy_Order_Details.csv

Purpose: Contains line-item details for each order (one order can have multiple items).

Column	Data Type	Description	Known Issues
Order ID	Text	Links to List of Orders	Some missing values
Amount	Text/Number	Sale amount per line item	Currency symbols (₹), comma decimals, text values

Column	Data Type	Description	Known Issues
Profit	Text/Number	Profit per line item	Parentheses for negatives, percentage signs, text
Quantity	Number	Units sold	Some negative values (data entry errors)
Category	Text	Product category	Inconsistent casing and abbreviations
Sub-Category	Text	Product sub-category	Typos, inconsistent naming
Discount	Text	Discount info	Mostly null, irrelevant

 **Data Quality Issues to Fix:** - Currency Symbols in Amount: Values like ₹1275.00, 632,0 (comma decimal), 2457.0 INR - Parentheses for Negative Numbers: Profit shows (1148.0) instead of -1148.0 - Percentage Signs: Some profit values have % appended - Missing Order IDs: Some rows have no Order ID (can't link to orders) - Negative Quantities: Data entry errors with negative quantities - Zero Amounts: Some rows have amount = 0 but positive profit - Inconsistent Categories: "Furniture" appears as "furniture", "FURNITURE", "Furn." - Sub-Category Typos: "Hankerchief" vs "Handkerchief", "Saree" vs "Sarees" - Blank Rows: Several completely empty rows at the end - Irrelevant Column: Discount column is mostly empty

2.3 Messy_Sales_Target.csv

Purpose: Contains monthly sales targets by product category.

Column	Data Type	Description	Known Issues
Month of Order Date	Text	Month and year	Multiple formats (Apr-18, April 2018, 4/2018, etc.)
Category	Text	Product category	Inconsistent casing, abbreviations
Target	Text/Number	Sales target amount	Currency symbols, commas, text placeholders
Region	Text	Region info	Mostly null, irrelevant

● Data Quality Issues to Fix: - Inconsistent Month Formats: Apr-18, April 2018, 4/2018, 7/2018, 1/2019 - Inconsistent Category Names: Mixed case and abbreviations - Currency Formatting in Target: Values like ₹9,000.00, Rs. 10,400.00, 9,000 - Text Placeholders: Some targets say “TBD” or “Pending” - Blank Rows: Empty rows with only Region filled - Swapped Columns: One row has Category and Month values swapped - Duplicate Rows: Several duplicate records - Irrelevant Column: Region is mostly empty

3. PART 1: IMPORTING DATA AS A FOLDER INTO POWER QUERY

Objective

Learn how to import multiple CSV files from a single folder using Power Query’s “From Folder” feature.

Background

When working with real-world data, you often receive multiple files that need to be processed together. Power Query's folder import feature allows you to: - Import all files from a folder automatically - Apply the same transformations to every file - Refresh the data when new files are added - Build a repeatable, automated pipeline

Instructions

Based on what you have learnt from the Study Group, if you have to visit the course, please do. This is a guide:

- Step 1: Organize Your Files
- Step 2: Open Power Query
- Step 3: Select the Folder
- Step 4: Choose Import Method
- Step 5: Configure the Combine Files Dialog
- Step 6: Understand What Power Query Creates

4. PART 2: DATA CLEANING & TRANSFORMATION QUESTIONS (Basic to Intermediate)

Objective

Clean and transform each dataset to resolve the data quality issues identified above.

SECTION A: Cleaning Messy_List_of_Orders.csv

Question 2.1: Remove Irrelevant Columns

Question 2.2: Remove Blank Rows

Question 2.3: Trim Whitespace from Text Columns

Question 2.4: Fix Leading Zeros in Order ID

Question 2.5: Standardize Date Formats

Question 2.6: Standardize State Names

Question 2.7: Standardize City Names

Question 2.8: Handle Null Values

Question 2.9: Remove Duplicates

SECTION B: Cleaning Messy_Order_Details.csv

Question 2.10: Clean the Amount Column

Question 2.11: Clean the Profit Column

Question 2.12: Fix Negative Quantities

Question 2.13: Standardize Category Names

Question 2.14: Standardize Sub-Category Names

Question 2.15: Remove Rows with Missing Order ID

Question 2.16: Handle Zero Amounts

SECTION C: Cleaning Messy_Sales_Target.csv

Question 2.17: Standardize Month Formats

Question 2.18: Clean Target Values

Question 2.19: Remove Duplicates and Blank Rows

5. PART 3: ADVANCED DATA CLEANING & TRANSFORMATION

Objective

Master advanced Power Query techniques including custom M code, conditional logic, merging queries, and building reusable transformations.

SECTION A: Advanced M Code & Custom Functions

Question 3.1: Create a Custom Function to Clean Currency

Question 3.2: Create a Conditional Column for Profit Status

Question 3.3: Create a Custom Column for Revenue per Unit

Question 3.4: Extract Year and Month from Order Date

SECTION B: Merging and Appending Queries

Question 3.5: Merge Orders with Order Details

Question 3.6: Aggregate Order Details by Order ID

Question 3.7: Merge Sales Targets with Actual Sales

SECTION C: Advanced Text Transformations

Question 3.8: Fuzzy Matching for State Names

Question 3.9: Create a Reference/Mapping Table

SECTION D: Performance and Best Practices

Question 3.10: Query Folding

Question 3.11: Query Dependencies and Organization

6. PART 4: VALIDATION & QUALITY CHECKS

Objective

Verify that your cleaned data matches the expected output and meets quality standards.

Question 4.1: Row Count Validation

Question 4.2: Data Type Validation

Question 4.3: Null Value Check

Question 4.4: Duplicate Check

Question 4.5: Referential Integrity Check

Question 4.6: Business Logic Validation

SUBMISSION CHECKLIST

Before submitting your case study, verify:

- ☐ All three files imported correctly
- ☐ Folder import method used for at least one dataset
- ☐ All blank rows removed
- ☐ All duplicates removed
- ☐ All text columns trimmed
- ☐ Date formats standardized
- ☐ State names standardized
- ☐ City names in proper case
- ☐ Amount and Profit columns cleaned of currency symbols
- ☐ Negative quantities fixed
- ☐ Category and Sub-Category names standardized
- ☐ Null values handled appropriately
- ☐ Data types set correctly for all columns
- ☐ Row counts match expected values
- ☐ No orphaned records in Order Details
- ☐ Queries are well-named and organized
- ☐ Applied Steps are documented and logical
- ☐ Final data loads successfully to Excel

ADDITIONAL RESOURCES

- Microsoft Power Query Documentation:
<https://docs.microsoft.com/en-us/power-query/>
- Power Query M Language Reference:
<https://docs.microsoft.com/en-us/powerquery-m/>

COACH FEEDBACKS

Good luck with your case study! Remember: the goal is not just to clean the data, but to build a repeatable, automated process that can handle future messy data with minimal effort.

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